The yearning for play: A scoping review for adapting mental health games meaningfully

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Abstract

Stress management remains a critical challenge for college students navigating the pressures of academic life and personal development. Gamification, the use of game design elements in non-game contexts, has shown promise in enhancing engagement and providing effective interventions for mental well-being. This scoping review explores gamified mental health applications designed for higher education, emphasizing the role of play theory in fostering resilience and stress reduction. By systematically analyzing existing literature, we address three key research questions: (1) Which game elements grounded in play theory are most commonly used in mental health applications? (2) What ethical and design challenges arise in their integration? (3) How do gamified apps foster playfulness to enhance stress management? Our findings highlight effective game mechanics, ethical considerations, and the potential of gamification to support sustainable mental health interventions. The review concludes with recommendations for designing ethical, user-centric gamified applications that align with play theory principles.

CCS Concepts

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Keywords

Gamification, mental health, higher education, stress management, scoping review, play theory, resilience, adult play, applications, games, college students

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1 Introduction

The transition to higher education represents a pivotal period characterized by heightened psychological stress. In 2024, 20.7% of college students reported severe psychological distress, while 39.2% identified stress as a major barrier to academic success. This phase often involves significant disruptions to routines, increased academic demands, and the assumption of new responsibilities that challenge both mental well-being and academic achievement.

Digital mental health applications have emerged as innovative tools to bridge gaps in traditional mental health care. Gamification, in particular, has gained traction for its ability to foster engagement through game mechanics such as progress tracking, narrative storytelling, and social cooperation. However, its effectiveness in supporting stress management within the context of higher education remains underexplored. Additionally, gaps exist in understanding why certain gamified interventions succeed while others fail to sustain user engagement.

This scoping review evaluates the use of gamification in mental health applications targeted at undergraduate students. Guided by established play theory frameworks, our review addresses three research questions:

RQ1: Which game elements grounded in current play theory are most commonly used in mental health applications for college students?

RQ2: What ethical and design challenges arise when integrating these elements?

RQ3: How do gamified mental health apps foster playfulness to influence stress management and resilience among undergraduate users?

Our review first contextualizes gamification within mental health interventions, then examines the current landscape of game-based interventions. After detailing our methodology, we present findings that analyze the characteristics of identified studies and the impact of game elements on stress management outcomes. Finally, we synthesize patterns and propose actionable recommendations for designing effective gamified mental health interventions tailored to college students.

2 Related Work

To help understand how these inter-workings are functioning within innate desire and why they bring such excitement to communities if done correctly, an understanding of the intersection of games, play, and playfulness needs to be established. Research has been started on attempting to draw a rational within the educational sector, attempting to explain why some individuals feel so motivated to learn in respect to learning with games. Very little work is done within the realm of mental health, not attempting to draw the parallels as to why we see such a spike in interest and effectiveness in games for mental health. This scoping review aims to evaluate mental health applications that have been embedded with game mechanics in them and draw the rational as to what is providing the best solutions in terms of care and why through the lens of play theory.

2.1 Games in Mental Health

Gamification in mental health applications leverages game mechanics to address critical challenges in stress management and resilience-building, creating engaging therapeutic contexts tailored to individual needs. Central to its effectiveness are intrinsic motivators, including autonomy, competence, and relatedness, as highlighted within Self-Determination Theory. These motivators are critical in fostering growth and development, enabling individuals to take meaningful strides toward well-being. They also intersect with play theory, which underscores the importance of designing play spaces that resonate with users' unique creative desires and needs for self-expression. For instance, autonomy is promoted when users are empowered to make choices in their therapeutic journey, competence is cultivated as they achieve measurable progress, and relatedness is strengthened through opportunities for social connection and interaction.

This is especially significant in mental health contexts, where traditional interventions often face challenges with adherence due to their lack of engagement or flexibility to meet individual needs. Gamified applications like SuperBetter exemplify this approach by integrating structured achievements that offer clear, measurable goals, role-playing elements that enable narrative self-expression, and interactive challenges that foster emotional regulation and self-improvement. By embedding these mechanics within the framework of play theory, such applications can transform struggles into expressive, imaginative play, allowing users to engage deeply and meaningfully with therapeutic techniques. Researchers and designers must prioritize these game mechanics, aligning them with intrinsic motivators to maximize the therapeutic potential of gamified systems and ensure interventions meet the diverse and nuanced needs of their users.

2.2 Play Theory Frameworks

Play theory provides a robust framework for understanding user engagement in gamified applications, particularly within mental health contexts. The concept of the "magic circle," originally proposed by Huizinga, emphasizes the creation of a psychological space for play where users can engage in dynamic, failure-tolerant environments to safely explore and refine coping strategies. Whitton builds on this foundation by suggesting that play experiences must be adaptable and individually meaningful, tailored to the diverse needs of users. This adaptability is especially crucial in mental health applications, where personalization can significantly enhance engagement and therapeutic outcomes.

It is essential to go beyond static frameworks and consider the dynamic nature of play as it applies to mental health contexts to deliver therapeutic tools. College students face diverse stressors, including academic pressures, social challenges, and personal development needs, each requiring tailored interventions. Research suggests that some stressors, such as academic deadlines, can promote growth and resilience when managed appropriately, while others, like social isolation, may lead to disruption and emotional distress. Dynamic play frameworks can adapt to these varying stressors by providing flexible, personalized environments that align with individual needs. By creating tailored experiences that balance challenge and support, gamified mental health tools can foster emotional regulation and resilience, addressing both the beneficial and detrimental aspects of stress. By altering our perception of game mechanics and recognizing them as more than simple added features, we can create a more nuanced approach to understanding how people interact with games. For example, mechanics such as adaptive difficulty can help reduce academic stress by providing achievable challenges that align with a user's current abilities, while collaborative tasks can address social stress by fostering connection and teamwork. Creating flexible and personalized play spaces rooted in play theory requires a deep understanding of how each individual is motivated and engages with different types of play. This adaptability allows gamified applications to address diverse stressors effectively, from time-bound academic pressures to the isolation associated with social challenges. Research highlights that when users encounter mechanics tailored to their needs—such as narrative-driven quests for introspection or competitive elements for goal-oriented individuals—the therapeutic potential of gamified applications significantly increases. This dynamic and user-centric focus transforms applications into powerful tools for stress management and emotional resilience. This adaptability ensures that users encounter experiences that resonate with their varied motivations. Applications have the power to be dynamic and personalized, which is why they hold such promise within this space. Lessons learned from the gaming space highlight the importance of incorporating elements such as adaptive difficulty, immersive narratives, and collaborative features to create tailored user experiences. These elements address diverse stressors by offering users flexibility and engagement in ways that static interventions cannot. For example, adaptive difficulty ensures that users remain challenged yet not overwhelmed, while immersive storytelling enables users to connect emotionally with their journey. Collaborative features, on the other hand, foster a sense of community and shared goals, which can be critical in addressing social stressors. By applying these lessons, designers can create gamified mental health applications that amplify therapeutic outcomes and align closely with the nuanced needs of stressed college students, ultimately making these tools more effective and impactful.

2.3 Toward Fluid Adaptability in Gamification

While gamification has shown significant promise in mental health applications, its current reliance on rigid game mechanics often prioritizes effectiveness over adaptability. Many systems focus on achieving measurable outcomes through structured achievements, leaderboards, or predefined challenges, which can inadvertently limit their ability to engage diverse users meaningfully. This emphasis on effectiveness often neglects the dynamic and fluid nature of play, which is essential for addressing the unique and evolving needs of individuals, particularly college students facing varied stressors.

Play theory suggests that engagement emerges from adaptable, user-centric experiences that allow for exploration, creativity, and emotional resonance. Fixed mechanics such as leaderboards might motivate extrinsically but fail to foster intrinsic engagement for users who do not find competition rewarding. Similarly, structured tasks may provide clarity but lack the flexibility needed to address personal stressors like social isolation or performance anxiety.

By reimagining gamification through the lens of fluid playfulness, designers can create systems that adapt dynamically to users' emotional and cognitive states. For example, instead of static progress bars, applications could implement narrative-driven experiences that shift based on user input, offering introspection and emotional support. Collaborative tasks might evolve into opportunities for social play, fostering a sense of belonging and reducing loneliness. Adapting game mechanics to prioritize the experiential and relational aspects of play ensures that interventions are not only effective but also meaningful and engaging for diverse users.

This shift toward fluid adaptability highlights the need for a balance between structured effectiveness and the playful dynamism that resonates with individual users. Gamification in mental health must move beyond rigid implementations to embrace play as a flexible and transformative tool, enabling interventions that are responsive, inclusive, and deeply aligned with the needs of stressed college students.

Gamification has become a valuable tool in mental health applications, leveraging game mechanics to create engaging and therapeutic experiences. Over the past decade, these applications have demonstrated significant potential in addressing diverse mental health challenges, particularly stress management. Successful implementations such as SuperBetter have shown how structured goals, immersive narratives, and interactive challenges can enhance user engagement, foster emotional resilience, and support self-improvement.

Research highlights that gamified interventions work by harnessing intrinsic motivators, including autonomy, competence, and relatedness, as outlined in Self-Determination Theory. These motivators are critical in sustaining engagement and encouraging adherence to therapeutic practices. For instance, adaptive difficulty mechanics provide challenges tailored to individual abilities, preventing frustration while promoting growth. Collaborative features address social stress by fostering connection and shared goals, while narrative-driven tasks enable introspection and personal reflection.

Despite these successes, the integration of gamification into mental health remains an evolving field. Challenges such as balancing ethical considerations, ensuring user privacy, and maintaining long-term engagement require further exploration. By reflecting on lessons learned from existing applications, designers and researchers can develop more dynamic, user-centric tools that align closely with the nuanced needs of diverse users, particularly college students managing stress. This section outlines the current achievements and limitations of gamification in mental health, setting the stage for future innovations in this promising domain.

Playfulness is distinct from play, functioning as a mindset or approach rather than a specific activity. While play refers to structured or unstructured activities with defined parameters, playfulness embodies an attitude of curiosity, creativity, and openness to exploration. This distinction is crucial in mental health applications, as fostering playfulness can enhance intrinsic motivation by allowing users to engage with therapeutic tools in ways that feel personally meaningful and dynamic. By incorporating elements that encourage spontaneity and adaptability—such as open-ended narratives, flexible challenges, and opportunities for creative expression—applications can better support long-term user retention and provide more impactful mental health interventions.

3 Methodology

3.1 Scoping Review

This scoping review investigates the intersection of gamification, mental health, and play theory by exploring how game mechanics address stress management in higher education. The methodology adheres to Arksey and O'Malley's six-stage framework, enhanced by Levac, Colquhoun, and O'Brien's recommendations for stakeholder engagement and iterative refinement. The review systematically maps the landscape of gamification in mental health by:

(1) **Defining the Research Questions:** The review centers on three key questions addressing the most effective game elements, ethical and design challenges, and the role of playfulness in fostering stress resilience among college students.

(2) **Identifying Relevant Studies:** Comprehensive searches were conducted across databases including PubMed, ACM Digital Library, and Scopus. Keywords such as "gamification," "mental health," "stress management," "play theory," and "higher education" were combined with Boolean operators to ensure inclusivity and relevance.
(3) **Study Selection:** A rigorous inclusion process was applied to identify studies that explicitly explored gamification mechanics, play theory frameworks, and their applications in non-clinical stress management for college students. Duplicates, non-peer-reviewed works, and unrelated studies were excluded to ensure relevance.
(4) **Charting the Data:** Data from the selected studies were systematically extracted and organized to evaluate game mechanics, their alignment with play theory, and their reported outcomes in stress management contexts.
(5) **Collating and Summarizing Results:** Findings were synthesized to highlight key trends, identify gaps in existing research, and outline actionable insights for gamified intervention design. The synthesis prioritizes user-centered design principles and adaptive play frameworks.
(6) **Consultation and Iterative Refinement:** Input from reviewers, including HCI experts and mental health professionals, was incorporated to refine the scope and applicability of the review's findings.

By following this structured approach, the review ensures a comprehensive and rigorous analysis of how gamification leverages play to address diverse stressors. It aims to inform HCI practices and mental health application design with evidence-based insights that emphasize adaptability, personalization, and engagement.

3.1.1 Search Strategy. Search terms included combinations of "gamification," "mental health," "stress management," "play theory," and "higher education." Filters limited results to peer-reviewed articles published in English between 2010 and 2025. Relevant databases included PubMed, ACM Digital Library, and Scopus, chosen for their alignment with HCI and mental health research. An initial search yielded 1,611 articles, which were screened using predefined inclusion criteria. These criteria required studies to explicitly explore gamification within mental health contexts, examine play theory as a framework, or evaluate game mechanics tailored to stress management. Exclusion criteria removed duplicates, non-peer-reviewed works, and articles without empirical or theoretical relevance. Following title and abstract screening, 311 articles were excluded, and full-text reviews identified 66 articles for the final analysis. This systematic approach ensured that the review remained focused on the intersection of gamification, play theory, and stress management for higher education.

Initial database searches yielded a substantial body of literature (n=1,083 in ACM Digital Library alone), requiring a systematic refinement process. Articles were selected based on the following hierarchical criteria:

(1) **Relevance to Research Questions:** Studies were prioritized if they explicitly addressed either gamification in mental health applications or play theory as a framework for adult engagement. Due to the limited direct research on play theory in mental health apps targeting college students, this review combined studies on gamification and mental health with those exploring play theory among adults. This approach ensured a comprehensive analysis, bridging gaps in existing literature and aligning with the scoping review's goal of identifying effective and theoretically grounded interventions.
(2) **Inclusion of Game Mechanics and Play Theory Frameworks:** Articles needed to evaluate specific game mechanics (e.g., leaderboards, narrative elements) or use play theory to frame their analysis of user engagement and therapeutic outcomes. Given the limited direct research connecting play theory to mental health apps for college students, this review drew from multiple domains. Studies on adult play theory provided insights into the psychological dynamics of playfulness, while research on gamification and games in mental health highlighted the practical implementation of game mechanics. Finally, findings on stress management in individuals were used to understand the therapeutic potential of gamified interventions, bridging these areas to form a cohesive analysis
(3) **Empirical or Theoretical Basis:** Only peer-reviewed studies providing empirical data, systematic analysis, or robust theoretical insights were included to ensure methodological rigor. Given the interdisciplinary nature of this scoping review, the studies drawn came from varied domains, including psychology, human-computer interaction (HCI), and game studies. This breadth was necessary to explore how play theory, traditionally researched outside of application development, intersects with gamification in mental health and stress management contexts. By integrating insights from these diverse areas, the review establishes a comprehensive understanding of how gamified systems can address stress among college students.
(4) **Population Focus:** Preference was given to studies targeting higher education populations or young adults facing stress-related challenges. Stress, being highly subjective and variable, required the inclusion of diverse perspectives and methodologies. This review acknowledges that stress can manifest in different forms, such as academic, social, or personal challenges, and is often influenced by individual perceptions and contexts. By addressing a broad spectrum of stress-related issues, the review ensures a nuanced understanding of how gamification and play theory can support diverse user needs effectively.

DB	N	Key Terms	Filters
ACM	1,083	gamif*, play theory, mental health	2010-24
IEEE	94	gamif*, stress, students	English
PubMed	13	gamification, stress mgmt	Peer-rev
T&F	25	mental health, play theory	2010-24
EBSCO	246	gamif*, wellness, students	English
SciDir	58	gamif*, stress, students	2010-24
Springer	92	gamif*, well-being, students	Peer-rev
Total	1611	Initial results	
Final	66	After screening	

Table 1: Search Strategy Overview

(5) **Publication Date and Language:** Articles published between 2010 and 2025 in English were included to ensure the review reflects current trends and research.

This systematic selection process ensured that the included studies were both relevant and of high quality, aligning with the scoping review's objectives to analyze how gamification leverages play theory to address diverse stressors effectively.

3.1.2 Search Strategy and Keyword Selection. Searches were completed in ACM DL, Scopus, Google Scholar, PubMed, IEEE Xplore, Taylor and Francis, EBSCO, ScienceDirect, and Springer in Fall 2024 and Winter 2025. The search strategy included the following keywords included:

- **Gamification:** "gamification," "game mechanics," "game-based design."
- **Mental Health Contexts:** "mental health," "stress management," "resilience-building," "emotional regulation."
- **Play Theory:** "play theory," "playfulness," "dynamic play," "magic circle," "adult play"
- **Target Population:** "higher education," "college students," "young adults," "undergraduate users."

Search strings were created for each database as illustrated in table 1 providing an overview of the results from each database.

Search queries incorporated Boolean operators and wildcards to ensure broad inclusion of relevant studies. Given the interdisciplinary nature of this review, flexibility was essential to accommodate research from diverse domains. Filters limited results to peer-reviewed articles published in English from January 2010 to November 2024 to capture recent advancements in gamification, play theory, and stress management research. Articles without explicit mentions of game mechanics or those not engaging with play theory as a conceptual framework were excluded during screening. This flexibility allowed for drawing parallels between seemingly disparate fields, bridging gaps to form a cohesive understanding of how gamification and play theory intersect in supporting stress management for college students.

3.1.3 Inclusion and Exclusion Criteria. Articles retrieved from the database search were screened for relevance based on title and abstract. Screening criteria included mentions of gamification, play theory, or specific play types related to "magic circle" theory in conjunction with stress management or mental health. Papers were excluded if they:

- Did not address either gamification or play theory.
- Lacked relevance to mental health or college students.
- Focused exclusively on theoretical discussions without empirical data.
- Were duplicates from multiple databases.

Two independent reviewers conducted the screening process, resolving conflicts through consensus. This step excluded 311 references, refining the dataset to studies directly relevant to the research questions [? ?].

3.2 Full-Text Screening and Inclusion

Criteria were established to include or exclude papers based on their relevance to the research questions and objectives. These criteria were designed to ensure methodological rigor and alignment with the scoping review's focus on gamification, play theory, and mental health in higher education.

Inclusion Criteria:

(1) **Relevance to Research Questions:** Studies explicitly addressing gamification, play theory, or their application to stress management in college students.
(2) **Population Focus:** Research involving higher education students or young adults experiencing stress-related challenges.
(3) **Publication Standards:** Peer-reviewed articles published in English between 2010 and 2025.
(4) **Theoretical or Empirical Basis:** Articles providing robust theoretical frameworks or empirical data on gamified interventions or play theory.
(5) **Game Mechanics:** Studies evaluating specific game mechanics (e.g., progress tracking, narrative elements) and their alignment with play theory.

Exclusion Criteria:

(1) **Irrelevant Topics:** Papers not focused on gamification, play theory, or mental health contexts.
(2) **Population Mismatch:** Studies focused on populations outside the target demographic (e.g., children or clinical settings).
(3) **Non-Peer-Reviewed Literature:** Exclusion of grey literature, conference abstracts, or opinion pieces.
(4) **Insufficient Methodology:** Articles lacking clear relevance to the research questions or methodological rigor.

This structured approach ensured a comprehensive and high-quality analysis, bridging diverse domains to provide insights into the intersection of gamification, play theory, and mental health interventions for stressed college students.

4 Results

A total of 66 studies met the inclusion criteria for this scoping review, revealing several recurring patterns in the design and implementation of gamified mental health applications for stressed college students. As noted, many interventions defaulted to extrinsic reward structures (e.g., badges, points, leaderboards) aligned with Self-Determination Theory (SDT). However, deeper exploration of the included studies also suggests that the eudaimonic (meaning) and hedonic (pleasure) experiences often leveraged in commercial gaming to produce more meaningful and long-lasting enjoyment are largely absent in current mental health apps. By contrast, the current state of most gamified mental health alternatives relies on heuristics (quick motivators) to implement coping and stress management techniques, ***often leading to steep decline rates*** and quick fixes that do not sustain the target audience.

4.1 Common Reliance on Heuristic Motivators

Across the reviewed interventions, heuristic motivators such as short-term streaks, collectible points, progress bars, and simple "reward loops" appeared in approximately 70% of studies. These quick-hit engagement strategies are known for generating initial user interest by offering rapid feedback and easy-to-grasp goals. In several trials, participants reported that these features made the apps more "game-like" and enjoyable at the outset, consistent with the "visceral appeal" commonly discussed in commercial game design literature [?]. Nonetheless, sustaining engagement remained a challenge: once novelty wore off, participants often ceased regular app use, echoing the broader critique that purely extrinsic gamification can lead to short-lived motivation gains.

Yet, in contrast, a much smaller subset of apps (fewer than 20% of the studies reviewed) tapped into eudaimonic motivators—those that foster deeper personal growth, emotional resonance, or profound sense-making. Commercial games frequently achieve eudaimonic engagement through narrative depth, moral or introspective choices, and empathetic social interactions [? ?]. However, among mental health apps designed for stress management, eudaimonic elements took the form of:

- Rich storytelling or guided narratives that reflect real-life stressors,
- Reflective prompts encouraging introspection about one's emotional states,
- Collaborative mini-quests that emphasize empathy and shared success rather than competition.

Only a handful of interventions embedded these narrative or interpersonal layers into their core functionality. Where they did appear (e.g., some versions of *SuperBetter* and *Happify*), participants reported stronger emotional bonding to the app and a sense that their engagement went beyond mere task completion. These findings align with research suggesting that meaningful game experiences can sustain user investment, prompting introspection and personal growth more effectively than short-term reward systems [?].

4.2 Tensions Between SDT and Playful Engagement

Across the reviewed studies, there was a notable trend to incorporate transactional interactions—leaderboards, fixed challenges, and reward-based feedback loops—to engage students in stress management tasks. This approach aligns with classical SDT literature, wherein autonomy, competence, and relatedness are used to enhance motivation. Yet, in approximately half of the studies, participants reported short-lived engagement once the novelty of badges or point-earning had diminished. These findings echo critiques raised by Chou (2019), who argues that such "actionable gamification" often becomes a surface-level intervention, neglecting the complex emotional processes inherent in mental health support.

The reliance on quick, heuristic mechanics instead of deeper, eudaimonic ones suggests a missed opportunity. Commercial games routinely blend fast feedback loops with thought-provoking content to draw players into sustained interaction and emotional attachment [?]. These loops are designed to give players the freedom to fail and master challenges in an evolving setting. In the mental health context, however, the prevailing approach is often risk-averse, keeping designs simple and clinically "safe" (e.g., daily checklists, streak tracking). While such minimalistic gamification can encourage immediate use—and may mimic certain looping elements in games—***it rarely provides the same freedom of failure and mastery*** found in gaming spaces. Consequently, it seldom becomes the catalyst for transformation that true play can offer [?], a phenomenon Whitton's framework identifies as crucial for emotional exploration and personal growth.

From a scoping review perspective, these results highlight the scarcity of mental health apps that dare to incorporate complex storytelling, moral dilemmas, or deeper cooperative elements. Such eudaimonic features are precisely what commercial games use to remain engaging long after the initial "fun" has passed—yet they remain underexplored in the domain of stress management apps for college students.

4.3 Interweaving Play Theory: From Transactional to Transformative

An important distinction in our analysis is how video games naturally motivate through intrinsic experiences of mastery, narrative immersion, and social connection, whereas many mental health apps reduce user interaction to linear tasks and badges. Przybylski and Rigby (2010) reinforce the significance of intrinsic motivators for genuine user involvement, yet only a minority of applications in our sample successfully integrated imaginative, narrative-driven components that facilitate deeper emotional engagement. For instance, studies referencing "pure play" integrated daily quests or evolving story arcs, effectively placing users in an exploratory role that recognized stress as a dynamic challenge rather than a static list of tasks.

Table 2 illustrates how common game mechanics (e.g., points, levels, narrative, avatars, social cooperation) align with

Whitton's framework, categorizing them by play type and their corresponding impact on stress relief. While structured elements like achievements, badges, and points were prevalent across interventions (more than 75% of the studies), their capacity to deliver long-term stress relief tended to be "moderate." By contrast, mechanics aligned with "pure play" (e.g., narrative, quests, social cooperation) consistently showed high potential for fostering emotional exploration and resilience.

Table 2: Extended Analysis of Game Elements in Mental Health Apps Through Whitton's Framework

Game Element	Play Type	Magic Circle Role	Stress Relief Impact	Example Apps
Streak Tracking	Structured Play	Serious Gaming (Game + Extrinsic)	Moderate	Headspace, Calm
Daily Check-ins	Exploration Play	Pure Play (Game + Playful + Intrinsic)	High	Habitica, Sanvello
Narrative Missions	Imagination Play	Pure Play (Game + Playful + Intrinsic)	High	MindShift CBT
Meditation Timer	Puzzle Play	Distraction (Playful + External)	Low–Moderate	Calm
Avatar Customization	Imagination Play	Playfulness (Playful + Intrinsic)	Moderate	Habitica, InnerHour
Social Support Quests	Performance Play	Pure Play (Game + Playful + Intrinsic)	High	Happify
Chatbot Interactions	Exploration Play	Serious Gaming (Game + Extrinsic)	Moderate	Woebot
Team Challenges	Structured Play	Extrinsic Play (Game + Playful + Extrinsic)	Moderate	MoodMission
Mindful Eating Quests	Exploration Play	Pure Play (Game + Playful + Intrinsic)	Moderate–High	Sanvello
Time Mgmt & Virtual Tree Growth	Puzzle Play	Distraction (Playful + External)	Low–Moderate	Forest

4.4 Whitton's Framework and the Shift to Adaptive, User-Centric Design

Whitton's play framework (2022) offers a nuanced way to interpret these findings: rather than viewing gamification as static extrinsic motivators, it positions play as a complex, adaptive process that can be harnessed for emotional regulation. Among the 66 studies, the handful that adopted user-driven storytelling, cooperative challenges, or adaptive feedback loops reported more profound shifts in stress perception. Users in these studies often cited feelings of autonomy and creativity, paralleling the "magic circle" emphasis on psychologically safe spaces.

Furthermore, several authors have acknowledged that by reframing stress as a playful quest rather than a clinical symptom, students were more inclined to persist with interventions beyond initial novelty. ***Applications that elicit these user-driven, play-oriented behaviors***—aligned with Whitton's "pure play" or "imagination play"—allowed participants to experience intrinsic satisfaction and emotional safety. Such qualities are often overlooked when apps rely on competition or extrinsic goals alone.

4.5 Reflections on Therapeutic Implications

The theoretical implications are twofold. First, while SDT-aligned extrinsic approaches might provide immediate engagement, they often fail to support the layered, iterative nature of emotional growth. Second, incorporating playfulness—in the sense of imaginative exploration, safe experimentation, and social connectedness—enables interventions to tap into a deeper, more sustainable form of user motivation. Ultimately, this finding suggests that designing mental health apps for stressed college students requires both acknowledging the initial draw of achievements and ***elevating*** playful, adaptive elements that align with the magic circle concept.

Taken together, these findings show that while many apps effectively incorporate heuristic motivators to capture user attention, few leverage the eudaimonic strategies known to sustain player engagement and foster internal growth. This gap underscores a need to look beyond superficial "points and badges" models and develop interventions that elicit personal meaning, empathic community-building, and ongoing self-reflection. Such an approach could bridge the well-documented drop-off in app usage over time, potentially leading to more robust mental health outcomes for stressed college students.

In summary, results from the 66 reviewed studies reveal a mixed picture: gamification has proven promising for short-term engagement, but the interventions with the most lasting impact on stress management and emotional resilience are those that move beyond superficial point systems. Instead, they draw from robust play theory—particularly Whitton's framework—to embed meaningful, adaptive play experiences that honor the complexity of stress and encourage intrinsic, ongoing user participation.

5 Discussion

This scoping review examined how gamification is used within mental health applications designed for stressed college students, focusing on whether current implementations align with deeper play theory principles. Three major findings emerged. First, most interventions rely heavily on extrinsic reward systems—points, badges, leaderboards—providing immediate but short-lived motivational benefits. Second, few applications meaningfully integrate eudaimonic or more profound play elements, such as narrative depth, moral choice, or collaborative quests that foster emotional resonance and long-term engagement. Finally, the limited subset of interventions that do use "pure play" or "imagination play" mechanics (as described by Whitton) reported greater potential for sustained stress relief and user satisfaction.

5.1 Addressing the Research Questions

RQ1 investigated which game elements grounded in current play theory are most commonly used. Results revealed a preference for transactional mechanics such as badges, streak tracking, and points, while deeper, intrinsically motivating elements (e.g., rich storytelling, cooperative scenarios) were comparatively rare. This imbalance underscores the predominant reliance on Self-Determination Theory (SDT)-based extrinsic motivators—an approach that can generate initial uptake yet may not harness the transformative potential of play.

RQ2 probed the ethical and design challenges in integrating these elements. Although few reviewed studies addressed ethical concerns explicitly, the findings suggest multiple design-level hurdles. For instance, overly competitive features (e.g., public leaderboards) can inadvertently exacerbate stress or stigmatize certain users, while risk-averse designs limit opportunities for the emotional exploration that is central to resilience-building. Data privacy and confidentiality—critical in mental health contexts—were also rarely discussed in detail. These gaps highlight the need for user-centric, empathetic design approaches that safeguard privacy and promote inclusive, non-judgmental environments.

RQ3 focused on how gamified apps foster playfulness to influence stress management and resilience. Evidence points to the importance of playful exploration that transcends mere point-collecting: those apps incorporating quest-based narratives, imaginative role-play, or safe spaces for "failure" and growth reported higher engagement and stronger indications of improved well-being. By framing stress not as a strict clinical symptom but as a dynamic challenge within a "magic circle" of play, students can experiment with coping strategies more freely, potentially reinforcing resilience over time.

5.2 Theoretical Implications

Taken together, these findings stress the limitations of purely SDT-driven gamification in mental health. While SDT remains vital for explaining intrinsic vs. extrinsic motivation, it does not fully capture the multidimensional nature of play in therapeutic contexts. Eudaimonic design elements—rooted in meaning, emotional resonance, and personal growth—are central to Whitton's vision of "pure play," yet remain underutilized. This gap between well-established commercial game design (which often embraces deeper narrative arcs and moral/emotional complexity) and clinically oriented apps (which tend to err on the side of simplicity) may explain why user engagement often drops post-novelty phase.

Furthermore, bridging Whitton's play theory with mental health interventions highlights the potential for self-exploration and creative autonomy. Apps that adapt difficulty dynamically, offer branching storylines, or encourage collective "quests" could emulate the robust engagement seen in AAA or indie video games. Such a shift requires rethinking gamification not just as a means to track behavior but as a holistic, playful experience that acknowledges emotional nuance.

5.3 Practical Design Implications

A key takeaway for designers is the need to balance heuristic and eudaimonic motivators. While short-term reward loops capture initial attention—a critical first step—sustained engagement arises from features that foster personal meaning, social connectedness, and adaptive challenge. Specifically:

- Narrative Depth: Integrate storylines where users confront relatable stressors in a "safe" fictional setting, tapping into imagination and empathy.
- Collaborative or Empathetic Play: Replace or supplement competitive leaderboards with cooperative mini-quests or peer support challenges, emphasizing collective growth over individual ranking.
- Reflective Prompts: Offer adaptive feedback loops prompting self-reflection and introspection, reinforcing autonomy and competence in a meaningful way.
- Safe Failure: Encourage iterative experimentation (e.g., "try a new coping approach, see how it feels, level up your character"), mirroring how games treat failure as a learning experience rather than a final verdict.

Implementing these strategies requires close collaboration among designers, mental health professionals, and user communities to ensure interventions remain both engaging and ethically sound.

5.4 Ethical Considerations

Although ethical implications were not extensively detailed in most studies, certain themes warrant attention. Privacy is paramount in mental health interventions, especially if leaderboards or social features disclose sensitive personal information. Additionally, competition-based reward mechanics may introduce pressure or fear of judgment, potentially undermining the supportive intent. Designing with user well-being at the forefront—by providing opt-outs for public sharing, ensuring confidentiality, and prioritizing inclusive, stigma-free features—is essential to make gamified stress management both effective and responsible.

5.5 Limitations of the Scoping Review

Several limitations should be acknowledged:

(1) Language and Publication Bias: This review included only English-language studies, potentially excluding relevant interventions or culturally adapted apps published in other languages.
(2) Varied Outcome Measures: Studies measured "stress," "anxiety," or "well-being" inconsistently, making direct comparisons of effectiveness challenging.
(3) Short Study Durations: Many interventions evaluated usage for only a few weeks, leaving open questions about long-term engagement and lasting resilience.
(4) Underreporting of Ethical Concerns: Because few articles addressed privacy, data security, or potential negative effects, the ethical landscape remains underexplored.
(5) Scoping Review Scope: By definition, a scoping review maps breadth rather than performing in-depth quality appraisals, so findings should be seen as indicative rather than conclusive.

5.6 Future Directions

Given the promising albeit underutilized role of eudaimonic design, future research might:

- Investigate Longitudinal Impact: Examine how narrative-driven or "pure play" mechanics shape stress resilience over semesters or academic years.
- Explore Personalized Play: Evaluate adaptive systems that tailor quest difficulty, narrative arcs, or social features to individual student needs and preferences.

- Assess Ethical Guidelines: Develop frameworks that address privacy, informed consent, and the potential for negative competition in mental health contexts.
- Cultural Adaptations: Expand beyond Western-centric interventions to validate whether certain eudaimonic or cooperative mechanics resonate differently across cultures.
- Comparative Effectiveness: Compare heuristic-only designs (points, badges, leaderboards) with combined eudaimonic–heuristic systems to quantify differences in engagement, stress reduction, and user satisfaction.

6 Conclusion

This scoping review identified persistent gaps in how gamification principles are applied to mental health applications for stressed college students. While most interventions integrated easily implementable, extrinsic reward mechanisms—such as points, badges, and leaderboards—these features alone often failed to sustain meaningful engagement. Instead, deeper play theory approaches, particularly those aligned with Whitton's framework, suggest that integrating more adaptive, imaginative, and collaborative design elements can foster emotional exploration, promote self-reflection, and potentially bolster resilience.

Despite promising examples of narrative-rich, quest-based, and socially cooperative tools, only a fraction of reviewed studies ventured beyond superficial gamification to leverage the full transformative potential of play. Moreover, ethical and design considerations, such as safeguarding user privacy and avoiding undue pressure through public leaderboards, remain underexplored. Future research should examine how tailored, user-centric interventions that blend both heuristic (quick motivators) and eudaimonic (meaning-focused) design might achieve deeper, more enduring stress reduction and enhanced well-being among undergraduates. By reconceptualizing gamification as a dynamic, play-driven process—rather than a simplistic points-based system—practitioners and researchers can more effectively harness technology to support the mental health needs of college students.

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References

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